

Term Project - CS329E

CS329 E - Elements of Data Analytics
(15 points)

1 Description

The goal of this project assignment is to develop a data analytics project using a real-world dataset. Students should select a public dataset, think about a research project, for example, t-test, two-sample hypothesis tests, regression models, classification, clustering, or any data analytic task, implement the data model (using libraries or without libraries), evaluate, and interpret the results. You can apply more than one data model.

2 Public Dataset List

Select a data set from the available public datasets.

You can find a list of such public data sets here:

<http://www.teymourian.de/public-data-sets-for-data-analytic-projects/>

- The data set must not be very Large.
- You can reuse one of the assignment data sets.

3 Define a Project (4 Points)

Define a data science research question based on the selected dataset. You can select one of the listed datasets or search on the web, and pick up one of the available public datasets.

You should describe the following items

1. What is your data set about?
2. Clean up your data and reduce it to no more than 2000 observations if your data set is very large.
3. What is exactly your research question? What do you want to learn from data? What is your learning model ,e.g., a Classification, Clustering, etc?
4. What is your current expectations about the results?
5. How do you want to evaluate your project? How to access the correctness of your model? How well would you expect that the model will work?

4 Implementation (6 Points)

- You need to implement your project in python. You are allowed to use any Machine Learning Library like scikit-learn¹. You are also allowed to implement your project without using any libraries.
- You can use Jupyter Notebooks to implement your project and provide your documentations.
- Your code should be completed and be compilable without any errors. We should be able to read your documents, and be able to run your project.
- Run your implementation (on your Laptop or on Cluster if data is large) and generate the results.
- Provide the interpretations of your results.
- What can you do to improve your results? Apply your ideas to improve your results.
- Provide any references that you have used.

5 Create Recorded Presentation Video of your Project (5 Points)

- Create a document or a presentation (You can create a presentation using your Jupyter notebook, or create a powerpoint presentation, or other formats) to describe your project and results
- Describe your code.
- Describe the results of your project in a professional way.
- You may want to use visualization diagrams and describe the results based on some diagrams - but having diagrams is not a MUST have to get the full credit.
- Describe the model and results of your project in a way that every person in this field can read, enjoy and understand.

Record your presentation in form of a video recording. Present your talk on your laptop computer and record it using a desktop recording software.

You should use your webcam so that we can identify who is presenter.

You have the following 3 options to create a presentation video:

- **Method 1:** You can use the Panopto Desktop Recorder. You can find this software and documentations here <https://www.panopto.com/>.
- **Method 2:** You can record your presentation using ZOOM client application, start your zoom client, start a zoom meeting (you will be the only participant), click on share your desktop, turn on camera and record your presentation, end the zoom session and you will get a MP4 video file of your presentation. Upload the file to Blackboard system

¹<https://scikit-learn.org/>

- **Method 3:** Alternatively, you can use any desktop recording software, generate a MP4 file and upload MP4 to blackboard, or upload it to some other cloud storage (like google Drive, DropBox) and share with us the URL Link of your presentation (submit the link in Blackboard).

Note 1: Your video presentation should be between 8 minutes to maximum 15 minutes long (Min 8 mi, Max 15 min)

Note 3: You should turn on your Web Camera so that we can see your face during the presentation, and be able to identify the presenter.

Grading will be based on quality of your presentation, and correct describing of algorithms or concepts.

5.1 Turnins

- Submit your Jupyter Notebook or Python code
- Submit your documentation (It can be part of your Jupyter Notebook or a separate file like PPT or Docs, or other formats)
- Submit your Video Presentation. Upload your video file to Canvas Panopto Video Gallery.

5.2 Grading Criteria

Your term project will be graded based on the following rubrics:

1. Correct description of the selected dataset.
2. Correctness of your data preparation and cleanup approach.
3. General Correctness of data analysis approach.
4. Correctness of your implementation and results.
5. Correctness of your interpretations of the results.
6. Providing all references.